

**Effectiveness of Traditional Educational Games Model
(TEGM) on Achievement in Mathematics of Middle Stage
Students of Public Schools of Fatehpur**

Synopsis of the Dissertation

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1.0 INTRODUCTION –

Education is a continuous process of learning and acquiring knowledge, skills, values, and habits, which fosters the holistic development (mental, physical, social, spiritual) of a human being. Education occurs through both formal and informal methods, making human beings capable of living in society and instilling a sense of morality and tolerance within them. Education is such a medium that transforms an individual into a moral, social, cultural, and religious figure.

Education is a dynamic and evolving process that aims to meet the changing needs of learners in today's fast-paced world. One of the key concerns in the current education system is the declining interest and low achievement in mathematics, especially among middle stage students in both public and private schools. Mathematics is a core subject that forms the foundation of logical thinking and problem-solving skills. However, the continued reliance on traditional lecture-based methods often results in rote memorization, lack of engagement, and poor conceptual understanding among learners.

Man is a social animal; he is always influenced by society and, in turn, influences society. Every individual is useful to society. For this, it is necessary for them to learn social rules, manners, behavior, ethics, and other socially useful aspects. For all this, it is essential that such arrangements are made right from childhood so that qualities can be developed within an individual that aid in their holistic development. In the present time, education is the only process through which the multi-faceted development of an individual can be achieved. The importance of ECCE in helping the holistic development of children has been recognized in Indian traditions, and significant weight has been given to the early years of children's lives in India's diverse cultural landscape. [Lakshmi 2025]

Education is a fundamental necessity for achieving full human potential, developing a just and equitable society, and promoting national development. Providing universal access to quality education is the key to India's continuous progress and economic development on the global stage in terms of social justice and equality, scientific advancement, national integration, and cultural preservation. Universal higher education is the appropriate medium through which the optimal development and enhancement of the country's rich talent and resources can be achieved for the well-being of the individual, society, nation, and the world. In the next decade, India will be the country with the youngest population in the world, and India's future will depend on providing these youth with higher-quality educational opportunities. [N.E.P. 2020]

Mathematics is an essential part of school education that plays a crucial role in developing logical reasoning, problem-solving, and analytical thinking in students. The school level is the stage where children either develop a greater inclination toward subjects like mathematics or begin to distance themselves from subjects like mathematics due to not receiving proper

teaching. This stage is highly essential because the foundation built during these years significantly impacts the students' future performance and their perspective toward math.

Children should be given opportunities in schools to learn about various problems of life and develop the ability to control their environment through the medium of education. The main objective of the school is to bring about some desirable changes in the behavior of children through the process of teaching and learning. 'Teaching' means 'to inspire to learn'. As long as a concept is not learned, it has not been taught, and there is no meaning to effective teaching if it does not lead to effective learning. Teaching is far more than just skillfully presenting prepared lessons or lectures. Knowledge cannot be transferred from one person to another like money. Knowledge cannot be acquired until the student is ready to accept it. In other words, where learning does not take place, teaching cannot exist. Therefore, it would not be wrong to say that teaching means causing learning to happen.

In the present times, it is being accepted in the field of education that teacher-centric videos are more effective compared to student-centric education. Despite this, most of the new educational experiments are carried out keeping in mind urban areas and technical resources, the implementation of which does not become possible in actual rural school classrooms. This very disparity highlights the need for research on teaching methodologies that are easily accessible and adaptable to local conditions.

Traditional educational game models are especially beneficial for such environments where access to digital resources is not possible; therefore, they can prove to be significantly useful for semi-urban districts like Fatehpur. **[Patil V. S. (2004)]**

However, mathematics is considered a difficult and fear-inducing subject in many rural and semi-urban areas. A traditional educational game model acts as an effective teaching method for a subject like math, which is considered to be a resource-scarce yet crucial component in mathematics education, particularly in districts like Fatehpur, where most schools are located in rural areas. Many challenges have to be faced in a resource-constrained classroom environment. Many classrooms operate with low technological support, limited infrastructure, and learners from diverse backgrounds. Students often come from such socio-economic conditions where academic support outside the school is limited. In such situations, the use of digital and modern equipment is not always possible. Therefore, there is a great need for such new, low-cost, and culturally integrated teaching methods that can improve learning without relying on advanced infrastructure. A traditional educational game model is a living example of this.

According to the National Curriculum Framework (NCF), the purpose of mathematics education is not merely to impart knowledge of numbers, formulas, and calculations, but rather to develop logical thinking, problem-solving abilities, and mathematical reasoning in students. The NCF believes that learning mathematics is an active intellectual process in which students reason, make assumptions, and clarify their thoughts explicitly. It has been clearly stated in the

NCF that mathematics should not be taught as a subject to be memorized, but should be taught through experiences connected to daily life. Daily life problems, activities, games, experiments, and discussion-based teaching methods make mathematics interesting and meaningful. This reduces fear of mathematics among students and develops self-confidence.

Additionally, the NCF also suggests that mathematical evaluation should emphasize projects, oral presentations, problem-analysis, and practical activities rather than being limited only to written examinations, so that a correct assessment of the students' thinking process can be made. In this way, NCF emphasizes making mathematics education understanding-based, experiential, and useful for life. **(NCF 2005)**

According to the National Policy on Education (NPE), 1986, the purpose of mathematics education is not only to develop quantitative efficiency but also to cultivate logical thinking, analytical capacity, and problem-solving skills in students. The policy clarifies that mathematics should be taught as a subject at the middle stage (Classes 6-8) that strengthens the children's ability to think, reason, and make decisions. **[National Policy on Education (NPE), 1986]**

NPE-1986 emphasizes that mathematics teaching should be shifted away from the rote learning system and made activity-based and connected to life, so that students' fear towards mathematics is reduced and their interest is developed. Along with this, it has also been mentioned in the policy that modern techniques and innovations should be utilized in mathematics teaching so that students can understand mathematical concepts more clearly. In this way, NPE-1986 emphasizes intellectual development and analytical capacity building at the middle stage through mathematics.

When these are used in mathematics teaching, complex and abstract concepts transform into concrete experiences. Students learn not just by listening, but by doing, seeing, and active participation. This viewpoint aligns with the active learning theory, which posits that knowledge is actively constructed by the learner through experiential and social engagement. Traditional educational game models are particularly useful for the middle school stage because they reduce several major barriers encountered in learning.

First of all, they create a joyful and supportive learning environment, reducing fear and anxiety toward mathematics. When presented in the form of a game, students' natural hesitation toward learning decreases. Secondly, it increases the active participation of students. In traditional classrooms, often only a few students respond while others remain silent. Game-based activities prompt every student to participate, fostering a spirit of cooperation and collaboration. Thirdly, this helps in retaining concepts in long-term memory because active engagement leads to deeper understanding and sustainable learning.

Natural numbers, whole numbers, integers, geometrical concepts like angles, triangles, and circles can be taught effectively through structured games using locally available resources

such as stones, ropes, sticks, chalk, clay, matchsticks, and other daily objects. These tools are low-cost and easily available in rural areas, making the teaching methodology practical and sustainable. They improve the retention of mathematical ideas through physical activity and visual representation, helping students shift from rote memorization to conceptual understanding.

This study focuses on analyzing the impact of a traditional educational game model on the mathematical achievement of middle school students in the Fatehpur district. Mathematical achievement here does not mean merely the ability to calculate, but also involves understanding concepts, applying them to new situations, and developing a positive attitude toward the subject. Academic achievement is considered a significant indicator of educational success, and its utilization is frequently used to test the effectiveness of teaching methods.

Wood (2013) presents some evidence regarding the importance of games. Their summary is that in psychoanalytic and therapeutic traditions, play is significant for healthy social and emotional development. It demonstrates that students experience themselves as autonomous individuals, building and managing their identity, and negotiating relationships with others in their context. It enables students to cope with life and develop strategies to adapt to its demands. Wood (2013) argues that in the field of cognitive development, it includes lateral thinking, adaptability, problem-formation, and problem-solving, along with symbolic activities, creativity, imagination, curiosity, and exploration.

Mathematical Achievement - Achievement refers to the successful achievement or performance gained on the basis of skill, hard work, and interest in a specific subject (Educational Dictionary, 2006). In mathematics, achievement means successful achievement or performance in math. For the present study, mathematical achievement is defined in the form of total marks obtained by students in a mathematics achievement test conducted by an individual.

Mathematics can be defined as a human-made discipline or system that includes the development and establishment of systems of rules, which operate within various types of problem-solving and other formal mathematical activities. Counting, calculating, and other formal mathematical activities are the result of acquired knowledge (Dehaene, 2011). Yet, it appears that our mathematical intuition and foundational processes of understanding are present universally across different cultures, embedded within human species (Dehaene, 2011; Feigenson et al., 2004). According to Dehaene (2011), individuals possess an inherent pre-verbal numerical module. Apart from general learning capacities, children are born with specific learning mechanisms, which are also called mental structures. The acquisition of mathematical skills in young children is noteworthy, and their pre-verbal understanding can be used as the foundation for pre-school mathematical development.

Using gaming strategies is a method of teaching. Research studies have been conducted for game strategies in language education (Erkir, 2015; Anastasia, 2012), health science (Pedal,

2011; Blakely, Skirton, Cooper, Elmes and Nelms, 2010; Blakely, et al., Nelms, 2009; Royse and Newton, 2007), and science education (Aina, 2013; Echeverri and Sadler, 2011). [Bragg (2012)] has prepared a method.

Gaming strategies as a teaching tool: Some studies show that the field of mathematics education gives priority to games for teaching (Rule and Forsyth, 2015; Ferguson, 2014; Buil et al., 2012; Novak, 2012; Vogel and others, 2006; Randel, Morris, Wetzel and Whitehill, 1992). Although achievement through the medium of games in these studies has been tracked, the impact of the Syntectics Model and Gaming Strategy on achievement and creativity in mathematics among secondary school students was determined by researchers, as the study has not been fully conducted by researchers yet.

In recent years, attention has been focused on the subject of early mathematical development and acquisition in children. The reason for this is that research shows that children's early understanding and competency in mathematics are deeply related to their future performance in this subject.

In this context, there is an urgent need for innovative, engaging, and culturally relevant teaching approaches. The **Traditional Educational Games Model (TEGM)** offers such a solution by integrating **indigenous games and play-based learning** into the mathematics curriculum. Games like hopscotch, stick games, number puzzles, board games, and others have long been a part of Indian culture and learning. These games are interactive, collaborative, and hands-on, which makes mathematical concepts more relatable, enjoyable, and easier to grasp.

Unlike modern digital tools, traditional games are **low-cost, easily accessible**, and can be implemented in both **resource-rich and resource-poor settings**. The district of Fatehpur, with its unique mix of rural and urban schools, provides an ideal environment to assess the effectiveness of **TEGM**. This study seeks to explore whether the **TEGM** approach enhances students' academic achievement in mathematics, fosters active engagement, and supports inclusive learning across different types of schools. The findings may contribute to the design of innovative and culturally grounded pedagogical strategies that improve mathematics education at the foundational level.

The **Traditional Educational Games Model (TEGM)** offers a promising, culturally grounded approach that integrates traditional games with mathematical learning. These games- such as board games, Number, puzzle , stick games and hopscotch - are interactive, collaborative and easy to implement. They make abstract concepts tangible, encourage problem-solving, and foster social and cognitive development. Unlike digital tools, traditional games require minimal resources, making them suitable for both well-equipped and under resourced schools.

Fatehpur, with its mix of **rural-urban educational landscape**, presents an ideal setting to examine the effectiveness of **TEGM** in both **public and private schools**. This study aims to

asses whether TEGM improves mathematics achievement among middle stage students compared to conventional methods. The finding can guide educators and policymakers toward adopting engaging, inclusive , and culturally relevant strategies to enhance foundational mathematics education.



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1.1 NEED & SIGNIFICANCE & JUSTIFICATION OF THE STUDY

- Mathematics is a core subject that significantly influences students' logical thinking and academic success. However, many middle-stage students show low achievement and disinterest, especially in traditional classrooms. Public schools often lack resources, while private schools may not use culturally engaging methods. The need arises for a low-cost, effective, and inclusive teaching approach. Traditional Educational Games (TEGM) offer a culturally relevant, interactive, and engaging method that may improve mathematical learning outcomes in both public and private school contexts.
- This study is justified by the growing gap in mathematics performance and engagement among middle-stage students, particularly in rural and semi-urban districts like Fatehpur.

While modern technological tools are not equally accessible, traditional educational games are culturally embedded, resource-light, and adaptable to all school types. By evaluating the effectiveness of TEGM, the study contributes valuable data to support innovative, inclusive, and pedagogically sound teaching practices. It also helps policymakers, educators, and curriculum planners make informed decisions about integrating traditional methods into mainstream mathematics instruction.

1.2 RATIONALE OF THE STUDY

- Despite the critical importance of mathematics in academic and real-life contexts, students at the middle stage often exhibit poor performance and low motivation in the subject. Conventional teaching methods, dominated by rote learning and textbook dependency, have limited effectiveness in developing conceptual clarity and interest, especially among students in public schools where resources are minimal. On the other hand, private schools, while relatively better equipped, also often overlook engaging, culturally-rooted strategies.
- Traditional Educational Games Model (TEGM) represents a promising pedagogical approach that integrates play, collaboration, and culture into learning. However, there is a lack of empirical research, especially in semi-urban and rural districts like Fatehpur, that evaluates the actual impact of TEGM on mathematical achievement in both public and private school settings. Therefore, this study is needed to address the research gap, validate the effectiveness of TEGM, and propose it as an accessible, inclusive, and meaningful teaching strategy

1.3 STATEMENT OF THE PROBLEM

**EFFECTIVENESS OF TRADITIONAL EDUCATIONAL GAMES MODEL (TEGM)
ON ACHIEVEMENT IN MATHEMATICS OF MIDDLE STAGE STUDENTS OF
PUBLIC SCHOOLS OF FATEHPUR**

1.4 OPERATIONAL DEFINITION OF THE KEY TERMS TRADITIONAL EDUCATIONAL GAMES MODEL (TEGM)

Operational Definition

—In this study, TEGM refers to the used of specific games (e.g. : hopscotch, board games, number-puzzles etc.) implemented over an academic term to teach mathematics to class 6 students .|

(ACHIEVEMENT)

Operational definition :-

— In this study achievement specifically refers to the mathematics performance scores obtained by Students in a researcher- developed pre-test and Post test.

These scores will be used to assess the effectiveness of the TEGM in enhancing students mathematical understanding and skills.¶

1.4 OBJECTIVES OF THE STUDY

- 1) To established the framework of TEGM model
- 2) To study the effect of TEGM on Achievement in Mathematics of middle stage students of public schools
- 3) To compare the effectiveness of Traditional Educational Games Model (TEGM) on Achievement in Mathematics of Middle Stage Students of public school by comparing pretest between control and experimental group
- 4) To compare effectiveness of Traditional Educational Games Model (TEGM) on Achievement in Mathematics of Middle Stage Students of Public Schools by pre-test and post-test within the experimental group.
- 5) To compare the effectiveness of Traditional Educational Games Model (TEGM) on Achievement in Mathematics of Middle Stage Students of public school by comparing posttest between control and experimental group

1.5 RESEARCH HYPOTHESES.

H₀₁ - There is no significant difference between the mean scores of Traditional Educational Games Model (TEGM) on Achievement in Mathematics of Middle Stage Students of public school by comparing pre-test between control and experimental group

H₀₂ - There is no significant difference between the mean score of Traditional Educational Games Model (TEGM) on Achievement in Mathematics of Middle Stage Students of Public Schools by pre-test and post-test within the experimental group.

H₀₃ - There is no significant difference between the mean score of Traditional Educational Games Model (TEGM) on Achievement in Mathematics of Middle Stage Students of public school by comparing post-test between control and experimental group.

1.6 DELIMITATION OF THE STUDY :-

1. The present study is confined to the middle stage students only. (Private and Public schools).
2. The area of the study includes Fatehpur District of Up only

3. The duration of intervention is the academic term
4. The study focuses solely on TEGM and does not include digital or commercial gamy.
5. The study is limited to middle stage students (6-8) in selected public and private schools of fatehpur
6. Only mathematics achievement in consideredll

2.0 REVIEW OF THE RELATED LITERATURE :-

- **Himmawan and Juandi (2023)**, in their study titled —Games-Based Learning in Mathematics Education: A Systematic Literature Reviewll, aimed to examine the effectiveness and trends of using games in teaching mathematics. Using the PRISMA method, they systematically analyzed 30 peer-reviewed studies published between 2018 and 2022. The selected studies involved students from primary to secondary levels across diverse educational contexts. The review concluded that game-based learning—especially traditional games—positively influences mathematical achievement, enhances motivation, and improves conceptual understanding. The authors emphasized the value of constructivist approaches and recommended the structured inclusion of educational games in mathematics instruction for better engagement and learning outcomes.
- **Richard Jallorina Taclay (2013)**, in his study titled —Effects of Mathematical Games Strategy on the Achievement of Students in High School Geometryll, investigated the effectiveness of using mathematical games as an instructional tool to enhance students‘ academic performance in Geometry. The primary objective of the study was to determine whether game-based strategies could improve students‘ understanding and achievement compared to traditional teaching methods. Using a quantitative experimental design, the study involved two groups of high school students—one taught using conventional methods and the other exposed to mathematical games integrated into the Geometry curriculum. The results showed that students in the experimental group performed significantly better in post-tests, indicating higher

retention, engagement, and comprehension of geometric concepts. Taclay concluded that mathematical games are a valuable educational strategy that not only boosts academic achievement but also fosters enthusiasm for learning, reduces mathematics anxiety, and encourages active student participation. His findings support the broader educational shift towards interactive and student-centered learning approaches in Mathematics classrooms.

- **Tanudidjaja, Pranata, and Bernadhed (2025)**, in their study titled —Game-Based Learning for Mathematics Lesson on 3rd Grade Elementary School, explored the integration of game-based learning approaches into elementary-level mathematics instruction. The objective of the study was to examine how the use of digital games could enhance student engagement, motivation, and learning outcomes in Mathematics, particularly in younger learners at the primary level. The researchers adopted a quasi-experimental research design involving third-grade students who were divided into a control group receiving traditional instruction and an experimental group engaged with educational games specifically designed for mathematics topics. The review of literature in their study highlighted several key themes from past research, including the importance of active learning in early education, the motivational benefits of gamified instruction, and the potential of educational technology to foster deeper understanding through visual and interactive means. The findings revealed that students who learned through game-based methods demonstrated improved problem-solving skills, better conceptual understanding, and greater enthusiasm toward mathematics lessons. The study concluded that game-based learning is an effective pedagogical strategy for elementary mathematics, offering both cognitive and affective benefits. It aligns with constructivist theories of learning and underscores the value of using developmentally appropriate digital tools to make learning more engaging and meaningful for young learners.

- **Meylani (2025)**, in the article titled —Gamification and Game-Based Learning in Mathematics Education for Advancing SDG 4: A Systematic Review and Qualitative Synthesis of Contemporary Research Literature^{ll}, conducted a comprehensive review to examine the role of game-based approaches in promoting quality education, aligning with Sustainable Development Goal 4 (SDG 4). The study aimed to synthesize recent global research on how gamification and game-based learning contribute to improving mathematical learning outcomes, equity, and inclusiveness in education. Employing a systematic review methodology, the researcher analyzed peer-reviewed literature from the past decade using qualitative synthesis techniques. The review identified recurring themes such as the positive effects of gamified learning environments on student motivation, engagement, and performance, particularly in Mathematics education across diverse educational contexts. The study emphasized that game-based methods promote active learning, facilitate better conceptual understanding, and foster collaborative problem-solving skills. The findings support the use of innovative, student-centered approaches as a pathway toward more equitable and quality education.

- **Stohlmann (2022)**, in the study titled —Two Modes of Game-Based Learning for Middle School Mathematics^{ll}, examined the effectiveness of both digital and physical game-based learning formats in enhancing mathematics education for middle school students. The objective was to analyze how different modes of games influence student engagement and learning outcomes in mathematical problem-solving. Through a qualitative and descriptive review of relevant literature and classroom practices, the study discussed how game-based learning strategies can increase motivation, support collaborative learning, and provide meaningful contexts for applying mathematical concepts. The literature reviewed indicated that both digital games and traditional physical games help reduce math anxiety and encourage persistence in problem-solving tasks. Stohlmann highlighted that while digital games offer interactivity and immediate feedback, physical games promote social interaction and peer-supported learning. The study concluded that both modes of game-based learning are effective in different ways and, when appropriately implemented, can significantly enrich the mathematics learning experience by making it more dynamic, student-centered, and enjoyable.

2.1 RESEARCH GAP

Despite growing recognition of the importance of innovative and student-centered pedagogies, limited research exists on the structured use of Traditional Educational Games Models (TEGM) in mainstream mathematics instruction at the middle stage, especially in the Indian context. Most existing studies focus on digital game-based learning, leaving a gap in understanding how culturally rooted, low-cost traditional games influence mathematical achievement and conceptual understanding.

Furthermore, there is insufficient comparative analysis between public and private schools, particularly in districts like Fatehpur, where disparities in resources and teaching methods persist. Previous research has not fully explored the operational frameworks, implementation strategies, or the long-term academic impacts of traditional game-based learning on diverse student populations.

This study aims to fill this gap by investigating the effectiveness of TEGM in enhancing mathematics achievement and engagement, and by comparing its impact across public and private schools. It will also contribute to developing a culturally relevant pedagogical model suitable for resource-constrained educational settings.

3.0 RESEARCH METHODOLOGY

3.1 APPROACH OF RESEARCH

- In this present study Quantitative type research will be used.
- The research approach to studying the effectiveness of Traditional Educational Games Model (TEGM) on mathematics achievement typically involves a quantitative methodology, using experimental or quasi-experimental design. This allows for a systematic comparison between control and experimental groups to examine how culturally-rooted, game-based strategies influence students' performance and engagement in mathematics across public and private schools.

3.2 TYPES OF THE RESEARCH - Experimental types of research will be used.

3.3 RESEARCH DESIGN

□ In this study, Quasi-Experimental Design (Non-Randomized Control Group Pre-test, Post-test Design) will be used in this study. The investigation was the trial base on the pre-test and post-test control group structure.

Design of the study

Groups	Pre Test(O_1)	Independent Variable (X)	Post Test(O_2)	Comparison
Experimental Group	M_{EGO_1}	Treatment	M_{EGO_2}	$M_{EGO_1} \sim M_{EGO_2}$
Control Group	M_{CGO_1}	Control Status	M_{CGO_2}	$M_{EGO_2} \sim M_{CGO_2}$
Between the Groups	M_{EGO_2}		M_{CGO_1}	$M_{EGO_2} \sim M_{CGO_1}$

MEGO1 = Mean of Experimental Group Observation 1

MEGO2 = Mean of Experimental Group Observation 2

MCGO1 = Mean of Control Group Observation 1

MCGO2 = Mean of Control Group Observation ²

3.4 VARIABLES OF THE STUDY

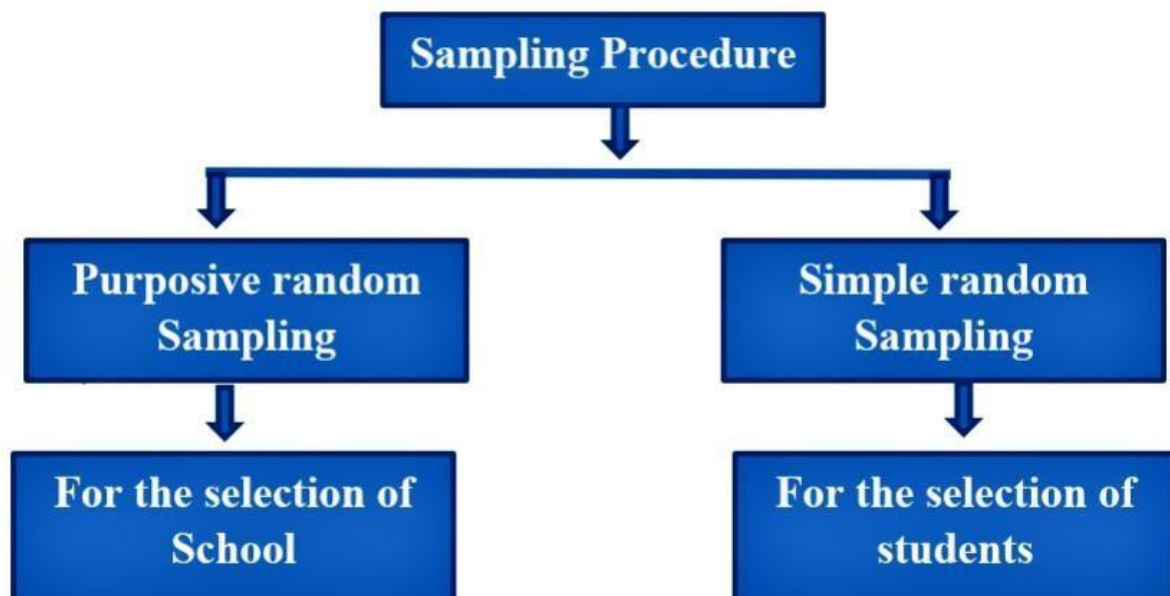
- **Independent Variable:**
Traditional Educational Games Model (TEGM)
- **Dependent Variable:**
Achievement in Mathematics
- **Control Variables:**
Grade level, syllabus content, duration of intervention

3.5 POPULATION OF THE STUDY

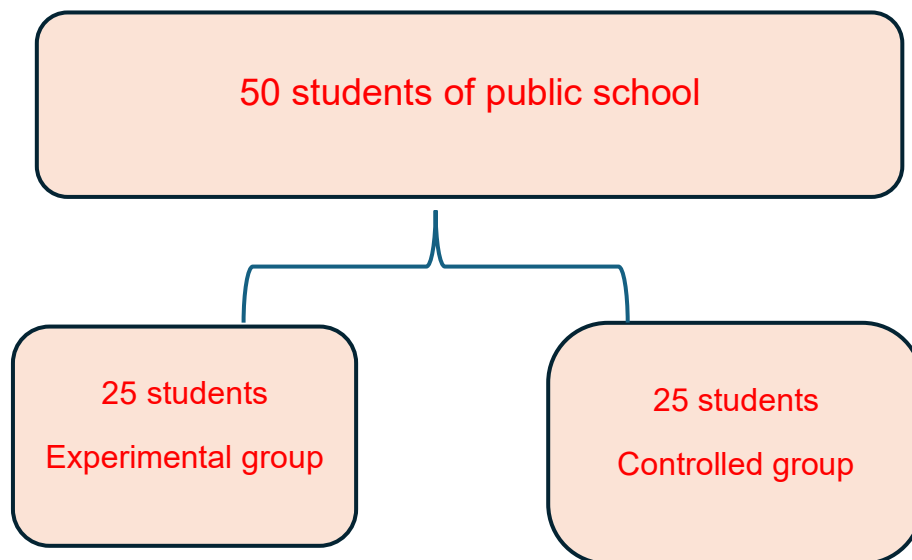
- All middle-stage (specially -Classes 6) students of public and private schools in Fatehpur district constitute the population. The focus will be on those schools which follow the same academic board (e.g., State Board or CBSE).

3.6 Sample and Sampling Techniques

The purposive sampling procedure will be utilized in the study for school selection and simple random sampling will be used in the selection for students.



Sample Size – For the purpose of the this study a sample size of 50 students will be selected.



3.6 TOOLS

□ Self constructed and validate achievement test of the mathematics.

3.7 DATA COLLECTION ANALYSIS AND INTERPRETATION

Appropriate statics techniques (Means,Standard Deviation , t-test, ANOVA ,Graphical representation) will be used for this type of study

3.8 EXPECTED OUTCOMES

After analyzing the data, the study will compare results between control and experimental groups to determine the effectiveness of TEGM. The discussion will interpret findings in the context of existing literature, explore how TEGM influenced learning, and explain why differences were observed. It will also highlight implications for teachers and curriculum designers, and recommend integrating traditional games into teaching practices, especially in low-resource schools.

4.0 PROBABLE CHAPTERIZATION

Chapter 1	Introduction
Chapter 2	Review of Related literature
Chapter 3	Research Methodology
Chapter 4	Probable Chapterization
Chapter 5	References

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